

1. Executive Summary

Metric	Value
Report Period	2026-08-31 - 2026-09-07
Total Threats Mitigated	79
High Priority	22
Low Priority	57
Manual Releases	0
Manual Blocks	0
ML Metrics	Value
False Positives	43
FP Rate	0.72%
IF Precision	99.89%
IF Recall	88.90%
IF F1-Score	94.07%
IF Accuracy	90.14%

Traffic Summary	Count
Total Flows Observed	49816
Total Packets Analyzed	269125863234
Malicious Packets	269123698223
True Negatives Passed	10774
Attack Vector	Count
ICMP Flood	18
SYN Flood	26
UDP Flood	18
Uncertain	17
Action	Count
Quarantined	10
Time Ban	10
Blackhole	3
Released	9

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.89%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	88.90%	Share of actual attacks successfully flagged
F1-Score	94.07%	Balanced measure combining Precision and Recall
Accuracy	90.14%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.72%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	11.10%	Actual attacks that went undetected
True Positive Rate (TPR)	88.90%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.28%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 38999	FN 4871
Actual: Normal	FP 43	TN 5903

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	21730	0	0
Act: ICMP	0	6660	0
Act: UDP	0	0	10609

2c. Response Latency

Metric	Value	Description
Detection Time	2879.9 ms	Average time taken to detect an attack
Mitigation Response Time	4.5 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.34%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	7.41%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	7.41%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	6	5	2026-09-07 14:06:18	2026-09-07 14:25:36
10.0.0.17	5	5	2026-09-07 14:05:46	2026-09-07 14:23:54
10.0.0.18	5	5	2026-09-07 14:05:47	2026-09-07 14:23:55
10.0.0.21	5	5	2026-09-07 14:05:53	2026-09-07 14:24:23
10.0.0.22	5	5	2026-09-07 14:05:48	2026-09-07 14:24:10
10.0.0.23	5	5	2026-09-07 14:07:16	2026-09-07 14:25:32
10.0.0.25	5	5	2026-09-07 14:05:50	2026-09-07 14:24:11
10.0.0.19	7	4	2026-09-07 14:06:16	2026-09-07 14:24:47
10.0.0.24	7	3	2026-09-07 14:06:08	2026-09-07 14:24:51
10.0.0.16	6	3	2026-09-07 14:06:17	2026-09-07 14:24:30
10.0.0.4	3	1	2026-09-07 14:25:02	2026-09-07 14:25:35
10.0.0.5	1	1	2026-09-07 14:25:06	2026-09-07 14:25:16

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 14:05:46	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:05:47	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:05:48	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:05:50	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:05:51	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-07 14:05:52	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-07 14:05:53	10.0.0.18	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-07 14:05:54	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:05:54	10.0.0.18	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-07 14:06:01	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-07 14:06:04	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-07 14:06:08	10.0.0.22	DDoS	UDP Flood	78.0%	High	Time Ban (1m)
2026-09-07 14:06:08	10.0.0.22	DDoS	UDP Flood	78.0%	High	Time Ban
2026-09-07 14:06:09	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:06:09	10.0.0.25	DDoS	ICMP Flood	90.3%	High	Time Ban (1m)
2026-09-07 14:06:09	10.0.0.25	DDoS	ICMP Flood	90.6%	High	Time Ban
2026-09-07 14:06:16	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:06:17	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:06:19	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:06:20	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-07 14:06:21	10.0.0.21	DDoS	UDP Flood	76.6%	High	Time Ban (1m)
2026-09-07 14:06:21	10.0.0.21	DDoS	UDP Flood	76.6%	High	Time Ban
2026-09-07 14:06:22	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-07 14:06:22	10.0.0.16	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-07 14:06:22	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-07 14:06:22	10.0.0.16	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-07 14:06:29	10.0.0.20	DDoS	Uncertain	-	Low	Released
2026-09-07 14:06:42	10.0.0.24	DDoS	ICMP Flood	91.8%	Critical	Time Ban (1m)
2026-09-07 14:06:42	10.0.0.24	DDoS	ICMP Flood	91.8%	Critical	Time Ban
2026-09-07 14:06:52	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-07 14:06:53	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-07 14:07:09	10.0.0.22	DDoS	UDP Flood	78.7%	High	Time Ban (2m)
2026-09-07 14:07:09	10.0.0.25	DDoS	ICMP Flood	91.9%	High	Time Ban (2m)
2026-09-07 14:07:16	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:07:22	10.0.0.21	DDoS	UDP Flood	76.6%	High	Time Ban (2m)
2026-09-07 14:07:23	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban (2m)
2026-09-07 14:07:23	10.0.0.16	DDoS	SYN Flood	94.1%	Critical	Time Ban (2m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 14:07:26	10.0.0.23	DDoS	Uncertain	-	Low	Released
2026-09-07 14:07:30	10.0.0.23	DDoS	ICMP Flood	89.4%	High	Time Ban (1m)
2026-09-07 14:07:30	10.0.0.23	DDoS	ICMP Flood	89.4%	High	Time Ban
2026-09-07 14:07:34	10.0.0.20	DDoS	UDP Flood	74.2%	High	Time Ban (1m)
2026-09-07 14:07:37	10.0.0.20	DDoS	UDP Flood	74.4%	High	Time Ban
2026-09-07 14:07:43	10.0.0.24	DDoS	ICMP Flood	92.2%	High	Time Ban (2m)
2026-09-07 14:08:30	10.0.0.23	DDoS	ICMP Flood	92.2%	High	Time Ban (2m)
2026-09-07 14:08:34	10.0.0.20	DDoS	UDP Flood	79.7%	Critical	Time Ban (2m)
2026-09-07 14:08:53	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-07 14:08:54	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-07 14:09:09	10.0.0.22	DDoS	UDP Flood	78.7%	High	Time Ban (5m)
2026-09-07 14:09:10	10.0.0.25	DDoS	ICMP Flood	91.9%	Critical	Time Ban (5m)
2026-09-07 14:09:22	10.0.0.21	DDoS	UDP Flood	76.6%	Critical	Time Ban (5m)
2026-09-07 14:09:23	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban (5m)
2026-09-07 14:09:23	10.0.0.16	DDoS	SYN Flood	94.1%	Critical	Time Ban (5m)
2026-09-07 14:09:44	10.0.0.24	DDoS	ICMP Flood	96.0%	High	Time Ban (5m)
2026-09-07 14:10:30	10.0.0.23	DDoS	ICMP Flood	92.2%	Critical	Time Ban (5m)
2026-09-07 14:10:34	10.0.0.20	DDoS	UDP Flood	79.7%	High	Time Ban (5m)
2026-09-07 14:13:53	10.0.0.17	DDoS	SYN Flood	93.5%	Critical	Time Ban (10m)
2026-09-07 14:13:54	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (10m)
2026-09-07 14:14:09	10.0.0.22	DDoS	UDP Flood	78.7%	High	Time Ban (10m)
2026-09-07 14:14:11	10.0.0.25	DDoS	ICMP Flood	91.9%	Critical	Time Ban (10m)
2026-09-07 14:14:23	10.0.0.21	DDoS	UDP Flood	76.6%	Critical	Time Ban (10m)
2026-09-07 14:14:24	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (10m)
2026-09-07 14:14:24	10.0.0.16	DDoS	SYN Flood	94.1%	Critical	Released
2026-09-07 14:14:29	10.0.0.16	DDoS	SYN Flood	94.1%	Critical	Time Ban (10m)
2026-09-07 14:14:50	10.0.0.24	DDoS	ICMP Flood	99.6%	Critical	Time Ban (10m)
2026-09-07 14:15:31	10.0.0.23	DDoS	ICMP Flood	92.2%	Critical	Time Ban (10m)
2026-09-07 14:15:35	10.0.0.20	DDoS	UDP Flood	79.7%	High	Time Ban (10m)
2026-09-07 14:23:54	10.0.0.17	DDoS	SYN Flood	93.5%	Critical	Time Ban (20m)
2026-09-07 14:23:55	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (20m)
2026-09-07 14:24:10	10.0.0.22	DDoS	UDP Flood	78.7%	High	Time Ban (20m)
2026-09-07 14:24:11	10.0.0.25	DDoS	ICMP Flood	91.9%	Critical	Time Ban (20m)
2026-09-07 14:24:23	10.0.0.21	DDoS	UDP Flood	76.6%	Critical	Time Ban (20m)
2026-09-07 14:24:24	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Released
2026-09-07 14:24:30	10.0.0.16	DDoS	SYN Flood	94.1%	Critical	Blackhole
2026-09-07 14:24:47	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Blackhole
2026-09-07 14:24:51	10.0.0.24	DDoS	ICMP Flood	99.6%	Critical	Blackhole

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 14:25:12	10.0.0.4	DDoS	Uncertain	58.8%	Low	Released
2026-09-07 14:25:16	10.0.0.5	DDoS	Uncertain	58.6%	Low	Released
2026-09-07 14:25:31	10.0.0.23	DDoS	ICMP Flood	92.2%	Critical	Time Ban (20m)
2026-09-07 14:25:36	10.0.0.20	DDoS	UDP Flood	79.7%	High	Time Ban (20m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6244	79.7%	High	Phase 2	5	18m 23s	2026-09-07 14:25:36	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6160	58.9%	Low	Phase 1	1	11s	2026-09-07 14:25:35	Attack Stopped
10.0.0.23	ICMP Flood	0.6496	92.2%	Critical	Phase 2	5	18m 2s	2026-09-07 14:25:32	Escalated to Time Ban
10.0.0.4	Uncertain	0.6121	58.9%	Low	Phase 1	1	11s	2026-09-07 14:25:24	Attack Stopped
10.0.0.5	Uncertain	0.6043	58.6%	Low	Phase 1	1	10s	2026-09-07 14:25:16	Attack Stopped
10.0.0.4	Uncertain	0.6205	58.8%	Low	Phase 1	1	10s	2026-09-07 14:25:12	Attack Stopped
10.0.0.24	ICMP Flood	0.5957	99.6%	Critical	Phase 2	1	10m 6s	2026-09-07 14:24:51	Ban Expired
10.0.0.19	SYN Flood	0.6395	93.1%	Critical	Phase 3	1	1s	2026-09-07 14:24:47	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6189	93.1%	Critical	Phase 1	2	6s	2026-09-07 14:24:32	Attack Stopped
10.0.0.16	SYN Flood	0.5698	94.1%	Critical	Phase 2	1	10m 6s	2026-09-07 14:24:30	Ban Expired
10.0.0.19	SYN Flood	0.5994	93.1%	Critical	Phase 2	1	18m 8s	2026-09-07 14:24:24	Ban Expired
10.0.0.21	UDP Flood	0.6318	76.6%	Critical	Phase 2	5	18m 2s	2026-09-07 14:24:23	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6496	91.9%	Critical	Phase 2	5	18m 2s	2026-09-07 14:24:11	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6307	78.7%	High	Phase 2	5	18m 22s	2026-09-07 14:24:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6148	93.1%	Critical	Phase 2	5	18m 8s	2026-09-07 14:23:55	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.5848	93.5%	Critical	Phase 2	5	18m 8s	2026-09-07 14:23:54	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6120	79.7%	High	Phase 2	4	8m 22s	2026-09-07 14:15:35	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6498	92.2%	Critical	Phase 2	4	8m 1s	2026-09-07 14:15:31	Escalated to Time Ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.24	ICMP Flood	0.6355	99.6%	Critical	Phase 2	3	5s	2026-09-07 14:14:50	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.5791	99.6%	High	Phase 2	1	8m 10s	2026-09-07 14:14:44	Ban Expired
10.0.0.16	SYN Flood	0.6253	94.1%	Critical	Phase 2	3	5s	2026-09-07 14:14:29	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6174	93.1%	Critical	Phase 2	4	8m 8s	2026-09-07 14:14:24	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6255	94.1%	Critical	Phase 2	1	8m 7s	2026-09-07 14:14:24	Ban Expired
10.0.0.21	UDP Flood	0.6322	76.6%	Critical	Phase 2	4	8m 2s	2026-09-07 14:14:23	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6498	91.9%	Critical	Phase 2	4	8m 2s	2026-09-07 14:14:11	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6266	78.7%	High	Phase 2	4	8m 21s	2026-09-07 14:14:09	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6324	93.1%	Critical	Phase 2	4	8m 7s	2026-09-07 14:13:54	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6100	93.5%	Critical	Phase 2	4	8m 7s	2026-09-07 14:13:53	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6125	79.7%	High	Phase 2	3	3m 22s	2026-09-07 14:10:35	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6502	92.2%	Critical	Phase 2	3	3m 0s	2026-09-07 14:10:30	Escalated to Time Ban
10.0.0.24	ICMP Flood	0.6036	96.0%	High	Phase 2	3	3m 10s	2026-09-07 14:09:44	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6150	91.0%	Critical	Phase 2	3	3m 7s	2026-09-07 14:09:23	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6338	94.1%	Critical	Phase 2	3	3m 6s	2026-09-07 14:09:23	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6303	76.6%	Critical	Phase 2	3	3m 1s	2026-09-07 14:09:22	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6502	91.9%	Critical	Phase 2	3	3m 1s	2026-09-07 14:09:10	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6284	78.7%	High	Phase 2	3	3m 21s	2026-09-07 14:09:09	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6373	93.1%	Critical	Phase 2	3	3m 7s	2026-09-07 14:08:54	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6198	93.4%	Critical	Phase 2	3	3m 7s	2026-09-07 14:08:53	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6346	79.7%	Critical	Phase 2	2	1m 21s	2026-09-07 14:08:34	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6388	92.2%	High	Phase 2	2	1m 0s	2026-09-07 14:08:30	Escalated to Time Ban
10.0.0.24	ICMP Flood	0.6419	92.2%	High	Phase 2	2	1m 9s	2026-09-07 14:07:43	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6332	74.2%	High	Phase 2	1	21s	2026-09-07 14:07:34	Attack persisted - escalating to time ban
10.0.0.23	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-07 14:07:26	Attack Stopped
10.0.0.19	SYN Flood	0.6253	91.0%	Critical	Phase 2	2	1m 7s	2026-09-07 14:07:23	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6365	94.1%	Critical	Phase 2	2	1m 6s	2026-09-07 14:07:23	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6176	76.6%	High	Phase 2	2	1m 1s	2026-09-07 14:07:22	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6323	78.7%	High	Phase 2	2	1m 21s	2026-09-07 14:07:09	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6350	91.9%	High	Phase 2	2	1m 0s	2026-09-07 14:07:09	Escalated to Time Ban
10.0.0.18	SYN Flood	0.6427	93.1%	Critical	Phase 2	2	1m 6s	2026-09-07 14:06:53	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6332	93.4%	Critical	Phase 2	2	1m 6s	2026-09-07 14:06:52	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6419	91.8%	Critical	Phase 2	1	8s	2026-09-07 14:06:42	Attack persisted - escalating to time ban
10.0.0.20	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:06:29	Attack Stopped
10.0.0.19	SYN Flood	0.6338	91.0%	Critical	Phase 2	1	6s	2026-09-07 14:06:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6421	90.9%	Critical	Phase 2	1	5s	2026-09-07 14:06:22	Attack persisted - escalating to time ban
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	12s	2026-09-07 14:06:20	Attack Stopped
10.0.0.22	UDP Flood	0.6359	78.0%	High	Phase 2	1	20s	2026-09-07 14:06:08	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:06:04	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:06:01	Attack Stopped
10.0.0.18	SYN Flood	0.6394	91.0%	Critical	Phase 2	1	6s	2026-09-07 14:05:53	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6369	93.4%	Critical	Phase 2	1	6s	2026-09-07 14:05:52	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE